# Amal S Pradeep

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## EDUCATION

**Northeastern University** | **Boston, MA** **Anticipated MAY 2027**

*Master of Science in Robotics - Relevant Coursework: Mechanics and Control, Control Systems, Sensors and Navigation, Mechatronics, Robot Design*

**National Institute of Technology, Trichy** | **Tiruchirappalli, India | (NIRF Rank 9)** **NOV 2020 - JUN 2024**

*Bachelor of Technology in Electronics & Communication Engineering | JEE Mains 99.3 percentile - Relevant Coursework: Embedded Systems, Microprocessors and Microcontrollers, Fundamentals of Automotive Systems, Digital Signal Processing, Machine Learning and AI*

## WORK EXPERIENCE

**Mechanical R&D Lead (Co-op), Archenaux Robotics (formerly Airnaux Robotics)** **MAY 2026 - AUG 2026**

- Led mechanical design of the company's V1 uncrewed underwater vehicle from concept through fabrication: chassis and structural layout, thrusters placed around a sealed electronics enclosure, and a hydrodynamic outer cover tuned for buoyancy and CB/CG alignment
- Directed a 3-person mechanical team (2 engineers plus a faculty advisor), releasing 30+ structural components and subassemblies in SolidWorks and Fusion 360, establishing the structural baseline carried into subsequent build cycles
- Ran flow and structural simulations to validate hydrodynamic loading and load paths ahead of fabrication, cutting design-iteration turnaround by over 60% versus prior review cycles and reducing downstream rework
- Developed a Pixhawk-based control architecture for manual and assisted operation, integrated with a custom electronics rail for serviceable mounting inside the sealed enclosure; designed a hull-adherence mechanism for stable surface attachment
- Owned BOM and manufacturing documentation, coordinating vendor procurement and fabrication schedules with electrical, controls, and software teams to reconcile structure against sensor placement and mounting envelopes

**Trainee Engineer, Levvan Engineering & Automation Pvt. Ltd** | **Chennai, India** **AUG 2024 - FEB 2025**

- Designed a complete 3D CAD model of an industrial tire-cutting machine in SolidWorks, optimizing the blade mechanism for cutting efficiency and manufacturability
- Programmed Siemens PLC ladder logic for an automated dual-tank pump control system with level sensing and fail-safe switching, removing manual intervention from tank changeover
- Developed and configured HMI interfaces for industrial automation systems, enabling real-time monitoring and control of production processes

**Industrial Manufacturing Intern, L&T Rubber Machinery** | **Kanchipuram, India** **JUL 2023 - AUG 2023**

- Analyzed tire building and curing machinery via CAD models and shop-floor observation, documenting how electrical, mechanical, and control subsystems interact across the production line
- Supported quality checks on hydraulic and pneumatic subsystems, verifying sensor and actuator function against acceptance criteria

## PROJECTS

**Hyperion: Local-LLM Fleet Operations Supervisor** | **Dell Pro Max GB10 Hackathon (4th Place)** **OCT 2026**

- Built the command-center UI for a 4-person team's operations supervisor running two local LLMs (Qwen3.6-35B on vLLM) on one Dell Pro Max GB10, monitoring a 20-robot Isaac Sim warehouse with zero cloud model calls
- Delivered a dependency-free web dashboard combining live robot camera feeds, fleet alerts, skill-gate commissioning charts, and scheduler views on a single screen

**Tracked Ground Robot - Novel Track Retracting Mechanism** **OCT 2025 - NOV 2025**

- Designed a multi-linkage track retraction mechanism for a tracked mobile robot, validating kinematics and range of motion in MATLAB before committing to CAD release
- Modeled the complete assembly and produced dimensioned engineering drawings for fabrication in SolidWorks

**WALL-E Inspired Mobile Robot Platform** **SEP 2025 - DEC 2025**

- Designed and modelled the robot chassis and drive mechanisms in SolidWorks, then FDM 3D printed and assembled the complete platform
- Developed a ROS2-based system for remote teleoperation, enabling real-time control and communication
- Integrated IMU and motor encoders to collect motion and odometry data, supporting basic mapping and localization

## TECHNICAL SKILLS

- **Design & CAD:** SolidWorks (part, assembly, drawing), Fusion 360, engineering drawings, BOM management, manufacturing documentation, design for manufacturability
- **Analysis & Simulation:** SolidWorks Flow Simulation, structural and hydrodynamic validation, MATLAB (kinematics), Proteus, LTspice, Modelsim
- **Fabrication & Hardware:** FDM 3D printing, mechanical assembly and integration, Pixhawk, Arduino, Siemens PLC and HMI, hydraulic and pneumatic systems
- **Software & Platforms:** Python, MATLAB, ROS2, Linux, Windows, GitHub, TensorFlow